

City Library Data Analysis Report

1. Project Overview

The goal of this project is to analyze City Library borrowing data and answer meaningful questions about borrowing behavior, popular books and authors, genre preferences, active members, and differences between member groups.

Business Questions

- How does borrowing activity change over time?
- Which books and authors are the most popular?
- Which genres are most popular, and how do preferences differ across grades?
- Who are the most active library members?
- How does borrowing behavior differ across grades, neighborhoods, and membership statuses?

2. Data Collection and Integration

The project combines data from multiple sources to build one dataset suitable for analysis.

- Member data from the database.
- Book and checkout data from the database.
- Detailed book information from a JSON file.
- Additional checkout records from an HTML source.

3. Data Exploration and Data Quality Assessment

The merged dataset initially contained 417 records. Before analysis, the dataset was explored to identify duplicates, missing values, inconsistent categories, incorrect data types, and logical inconsistencies.

3.1 Duplicate Records

A duplicate check identified 8 duplicated records. Duplicate records can inflate borrowing counts and produce misleading analysis results, so they were removed.

Key Insight: Removing duplicates improves the reliability of all borrowing-frequency calculations.

3.2 Missing Values

Missing values were found in several columns:

- first_name: 5 missing values
- last_name: 5 missing values
- grade: 41 missing values
- join_date: 11 missing values
- checkout_id: 26 missing values
- return_date: 91 missing values
- membership_status: 5 missing values
- neighborhood: 5 missing values

- `publication_year`: 35 missing values

Records with missing member names could not be reliably associated with complete member information, so those incomplete member-name records were removed. Other missing values were handled according to their analytical importance; for example, grade was filled using the median where appropriate.

Missing return dates were not automatically treated as errors because they may represent books that had not yet been returned or incomplete operational records.

3.3 Inconsistent Categorical Values

The exploration found inconsistent capitalization in categorical columns. For example, neighborhoods appeared as values such as Maadi, Maadi with different capitalization, and similar variations. Membership status also appeared as Active/active and Inactive/inactive.

These values were standardized so that the same category would not be counted as multiple different groups.

3.4 Incorrect Data Types

Date columns were converted to datetime format to support time-based analysis. Identifier columns such as `member_id`, `checkout_id`, and `book_id` were converted to string because they represent identifiers rather than numerical measurements.

3.5 Logical Data Inconsistency

A logical consistency check found records where the checkout date occurred before the recorded member join date. This is not logically consistent under normal library operations.

The issue was documented rather than arbitrarily corrected because there was not enough information to determine which date was incorrect.

4. Data Cleaning Summary

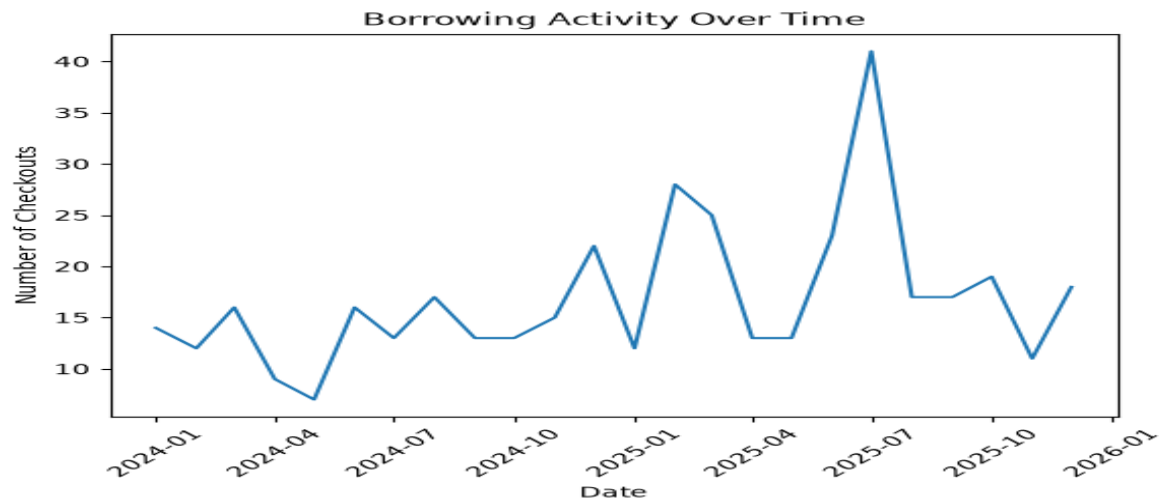
After the quality assessment, the following actions were completed:

- Removed duplicate records.
- Removed records with unrecoverable missing member names.
- Handled missing grade values using the median where appropriate.
- Converted date columns to datetime.
- Converted identifier columns to string.
- Standardized neighborhood values.
- Standardized membership status values.
- Documented the checkout-date-before-join-date inconsistency as a data limitation.

After cleaning, the final dataset used for analysis contained 404 records.

5. Analysis and Visualization

5.1 How Does Borrowing Activity Change Over Time?

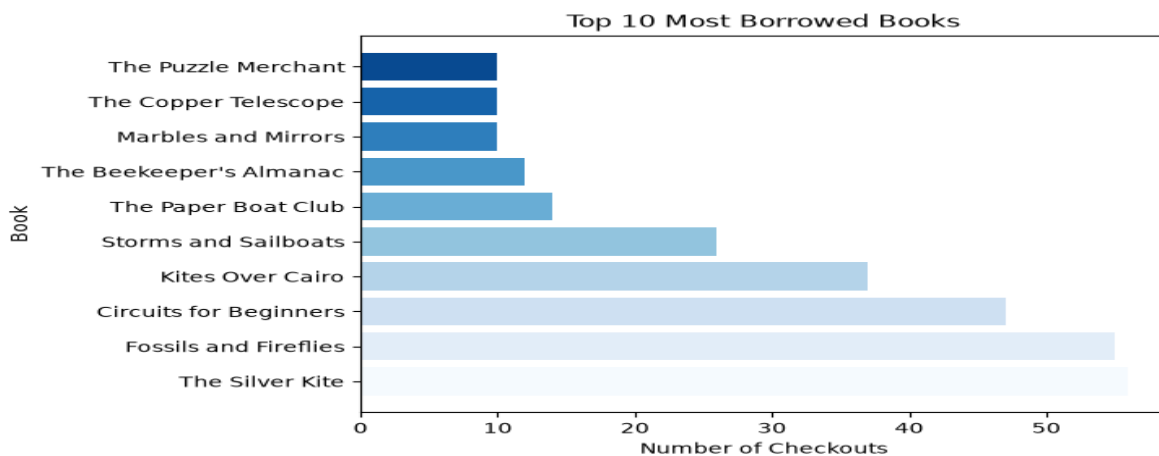


Summary: Borrowing activity changed across the observed period rather than remaining constant. The lowest recorded monthly activity was 7 borrowings in May 2024, while borrowing increased substantially during 2025 and reached a peak of 41 borrowings in July 2025.

Key Insight: Borrowing demand appears to have seasonal variation, with July 2025 showing the strongest activity.

Recommendation: The library should prepare for high-demand periods by ensuring sufficient book availability and staffing during months with historically higher borrowing activity.

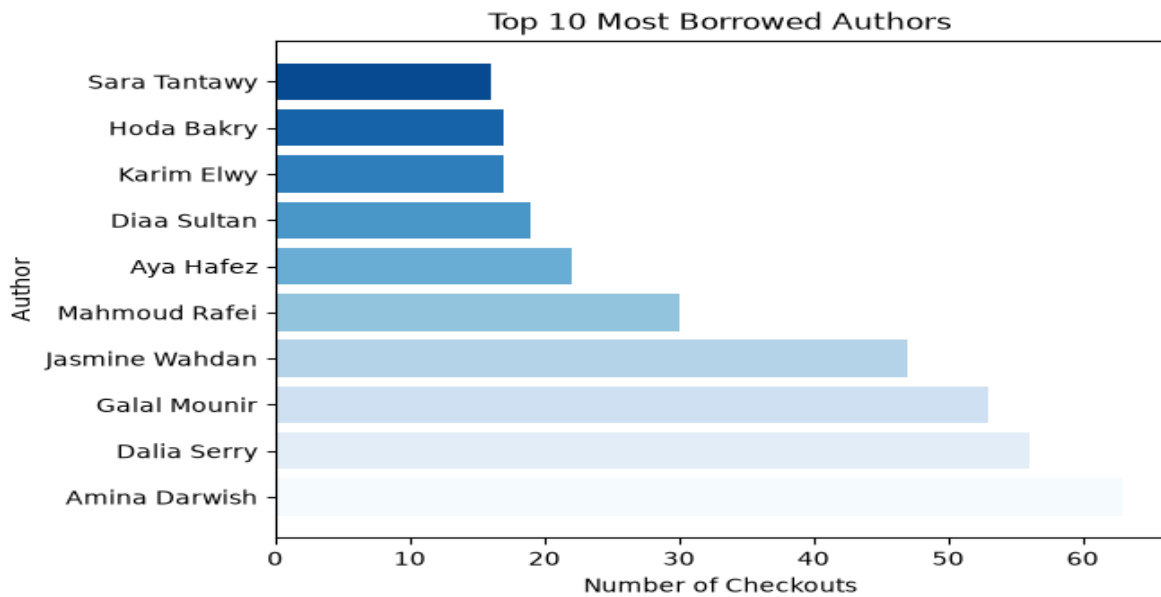
5.2.1 Which Books Are the Most Popular?



Summary: The Silver Kite was the most borrowed book with 56 borrowings, followed closely by Fossils and Fireflies with 55. Circuits for Beginners ranked next with 47 borrowings, followed by Kites Over Cairo with 37.

Recommendation: Monitor high-demand titles and consider maintaining additional copies to reduce potential availability problems.

5.2.2 Which Authors Are the Most Popular?

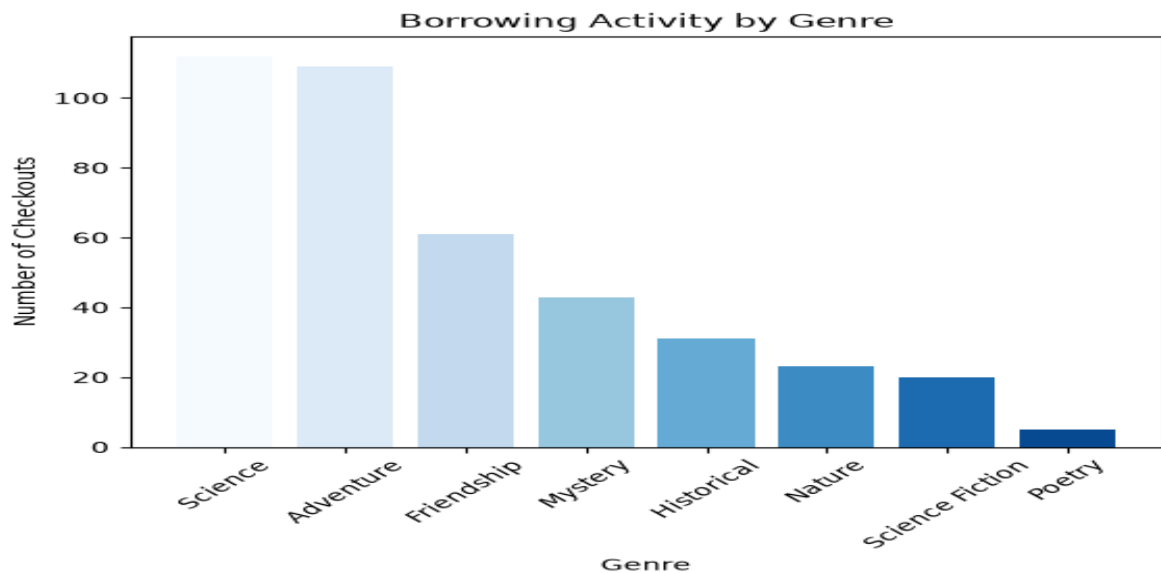


Summary: Amina Darwish was the most popular author with 63 borrowings, followed by Dalia Serry with 56 and Galal Mounir with 53. Jasmine Wahdan also showed strong demand with 47 borrowings.

Key Insight: Author popularity can help the library understand reader preferences beyond individual book titles.

Recommendation: Use author-level demand when planning future acquisitions and recommendations.

5.3.1 Which Genres Are Most Popular?

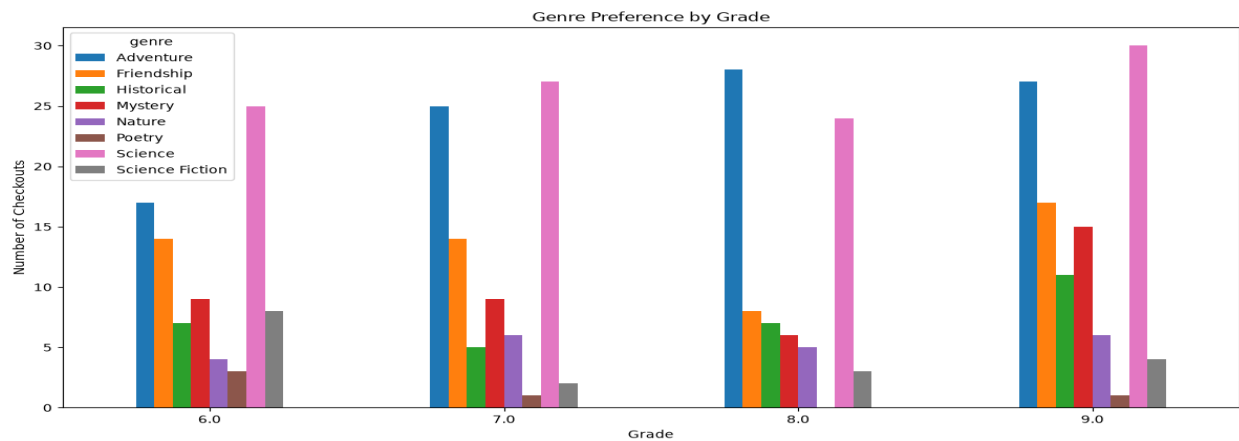


Summary: Science was the most popular genre with 112 borrowings, closely followed by Adventure with 109. Friendship ranked third with 61, while Poetry had the lowest borrowing activity with 5.

Key Insight: Science and Adventure are the strongest demand categories in the library.

Recommendation: Prioritize Science and Adventure when purchasing new books and reviewing inventory levels.

5.3.2 How Do Genre Preferences Differ Across Grades?

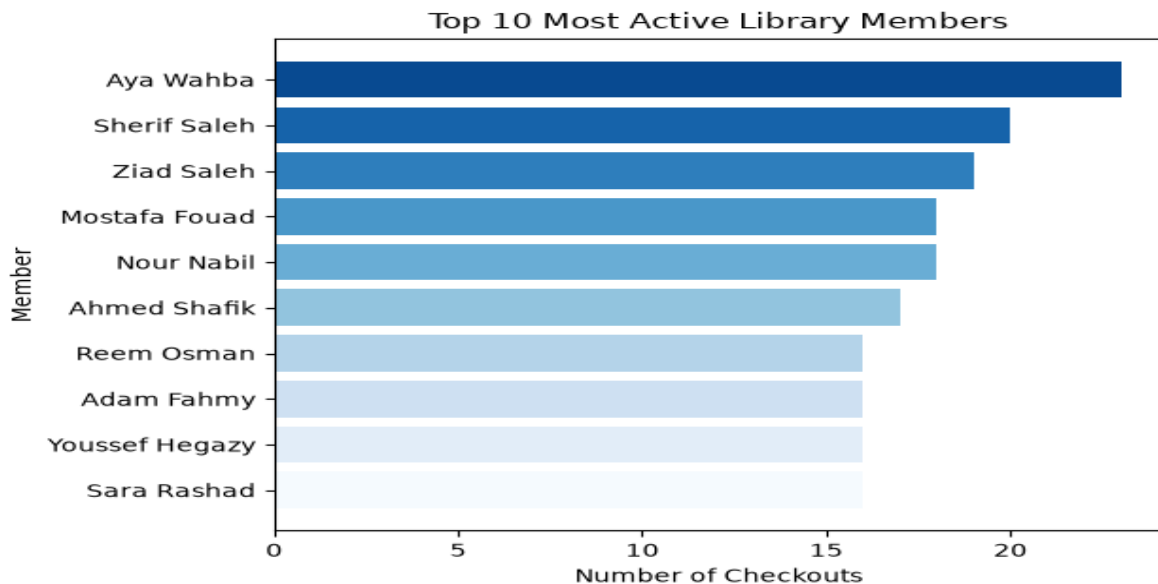


Summary: Genre preferences differed across grades, but Science and Adventure remained consistently strong. Grade 9 showed particularly high activity in Science, while Adventure was also popular across Grades 6–9.

Key Insight: Borrowing preferences can be analyzed by grade to support more targeted recommendations.

Recommendation: Create grade-based book recommendations and displays rather than using the same recommendations for all students.

5.4 Who Are the Most Active Library Members?

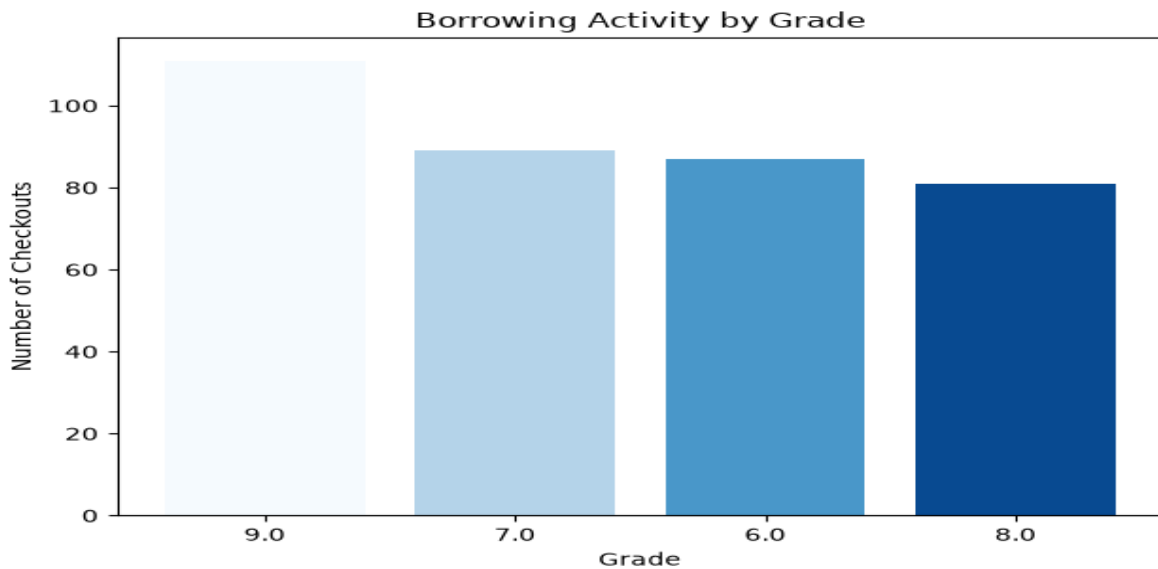


Summary: The visualization identifies the members with the highest borrowing frequency. The most active members demonstrate strong engagement with the library and represent the highest-frequency borrowers in the dataset.

Key Insight: A small group of highly engaged members can be useful for understanding successful engagement patterns.

Recommendation: Consider loyalty initiatives or personalized recommendations to maintain engagement among highly active readers.

5.5.1 How Does Borrowing Differ Across Grades?

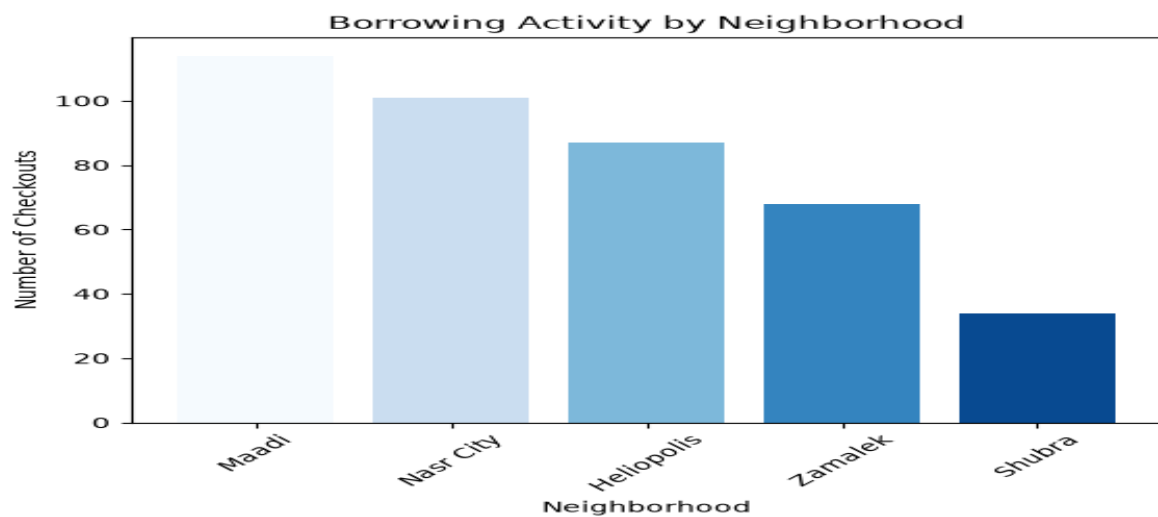


Summary: Grade 9 had the highest borrowing activity with 111 borrowings. Grade 7 followed with 89, Grade 6 had 87, and Grade 8 had 81.

Key Insight: Grade 9 students were the most active borrowing group in the final dataset.

Recommendation: Investigate what drives stronger engagement among Grade 9 students and apply successful practices to lower-engagement groups.

5.5.2 How Does Borrowing Differ Across Neighborhoods?



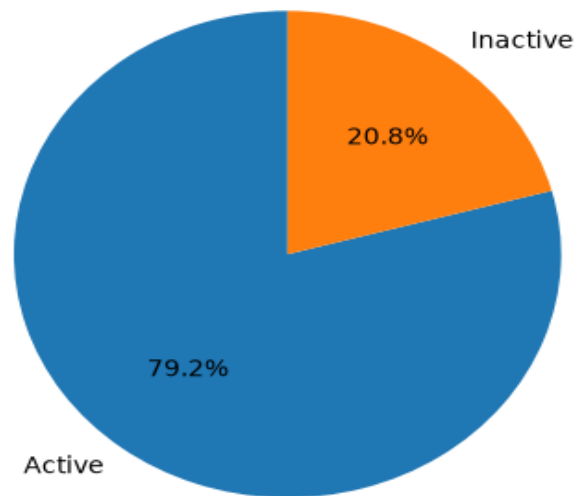
Summary: Maadi had the highest borrowing activity with 114 borrowings, followed by Nasr City with 101 and Heliopolis with 87. Shubra had the lowest activity with 34.

Key Insight: Borrowing engagement varies substantially by neighborhood.

Recommendation: Investigate lower activity in Shubra to understand whether awareness, accessibility, or member engagement contributes to the difference.

5.5.3 How Does Borrowing Differ by Membership Status?

Borrowing Activity by Membership Status



Summary: Active members accounted for 320 borrowings, compared with 84 borrowings from inactive members.

Key Insight: The majority of borrowing activity comes from active members.

Recommendation: Maintain active-member engagement while investigating whether inactive members can be re-engaged through reminders, recommendations, or membership renewal initiatives.

6. Overall Findings

- Borrowing activity varies over time and reached its strongest recorded level in July 2025.
- The Silver Kite and Fossils and Fireflies were the most frequently borrowed books.
- Amina Darwish was the most popular author.
- Science and Adventure were the most popular genres.
- Genre preferences showed differences across grades, while Science and Adventure remained consistently strong.
- Grade 9 had the highest borrowing activity.
- Maadi was the neighborhood with the highest borrowing activity, while Shubra had the lowest.
- Active members generated most borrowing activity.
- The dataset contained important quality issues, including missing values, duplicates, inconsistent categories, and logically inconsistent dates.

7. Final Recommendations

- Increase attention to high-demand Science and Adventure books.
- Monitor popular titles and consider additional copies where demand is consistently high.
- Use author and genre popularity to support acquisition decisions.
- Create grade-based recommendations to improve personalization.
- Prepare inventory for periods of higher borrowing demand.
- Investigate low borrowing activity in neighborhoods with weaker engagement.
- Use engagement initiatives to retain active members and potentially reactivate inactive members.
- Improve future data collection by applying validation rules to prevent checkout dates earlier than join dates and reduce missing operational data.